

5-Day NetDevOps Training Curriculum (DevOps Perspective)

Audience: Network Engineers, Cisco Engineers, Infrastructure Engineers

Focus: DevOps Practices Applied to Networking (NetDevOps)

Delivery Mode: Instructor-led with Hands-on Labs

Outcome: Participants will be able to automate, version, test, deploy, and validate network changes using DevOps methodologies.

Day 1 – DevOps Fundamentals for Networking

Objective:

Transition from traditional CLI-driven networking to a DevOps and automation-first mindset.

Topics Covered

- DevOps principles and culture for infrastructure teams
- Traditional networking vs NetDevOps
- Infrastructure as Code (IaC) for network devices
- GitOps concepts and change management
- Day 0, Day 1, and Day 2 network operations
- NetDevOps reference architecture

Hands-on Labs

- Linux basics: SSH, directories, permissions
- Network troubleshooting commands: ip, ss, tcpdump
- Git repository creation and versioning of network configurations

Day Outcome:

Participants understand the DevOps mindset and how Git becomes the single source of truth for network changes.

Day 2 – Python and APIs for Network Automation

Objective:

Use Python and APIs to automate repetitive and error-prone network tasks.

Topics Covered

- Python fundamentals for automation
- Working with JSON, YAML, and XML
- Network automation libraries: Netmiko, Requests, NAPALM (overview)
- Understanding REST APIs
- Cisco automation interfaces: REST API, RESTCONF, NETCONF, NX-API
- Orchestrating Network Services with Cisco NSO
- Network Automation Testing with pyATS

Hands-on Labs

- Python scripts to connect to Cisco devices
- Automated backup of running configurations
- Retrieve interface and device information using APIs
- Push configuration changes programmatically

Day Outcome:

Participants can build Python-based automation scripts and interact with network devices using APIs.

Day 3 – Ansible for Network Configuration Management

Objective:

Manage and standardize network configurations at scale using Ansible.

Topics Covered

- Ansible architecture and components
- Inventory and variable management
- Idempotency and desired state configuration
- Ansible network modules: ios_config, ios_command, nxos_config
- Secrets handling using Ansible Vault

Hands-on Labs

- Ansible playbooks for VLAN and interface configuration
- Standard baseline configuration deployment
- Configuration compliance validation
- Automated device onboarding

Day Outcome:

Participants can deploy, manage, and validate network configurations using Ansible playbooks.

Day 4 – CI/CD and GitOps for Networking

Objective:

Introduce CI/CD pipelines to safely test and deploy network changes.

Topics Covered

- CI/CD concepts tailored for networking
- GitOps workflow with pull requests and approvals
- Automated validation and testing before deployment
- CI/CD tools overview: GitHub Actions, GitLab CI
- Rollback and failure handling strategies

Hands-on Labs

- Create a Git-based network configuration workflow
- Build a simple CI pipeline for syntax and config validation

- Deploy network changes automatically using Ansible
- Simulate and recover from failed network changes

Day Outcome:

Participants can implement a CI/CD pipeline to deploy network changes reliably and safely.

Day 5 – Cloud Networking, Observability, and Capstone

Objective:

Apply NetDevOps concepts to cloud and production-like environments.

Topics Covered

- Cloud networking fundamentals (AWS/Azure - DevOps view)
- VPCs, subnets, routing, and security concepts
- Hybrid networking overview
- Monitoring and observability in NetDevOps
- Network telemetry and drift detection

Hands-on Labs

- Automate basic cloud network components
- Pre-change and post-change validation workflows
- End-to-end NetDevOps mini capstone project

Day Outcome:

Participants gain end-to-end visibility into NetDevOps workflows and real-world enterprise use cases.

Tools and Technologies Covered

- Linux
- Git & GitHub
- Python
- Ansible
- Cisco APIs (REST, RESTCONF, NETCONF)
- CI/CD Pipelines
- Cloud Networking Basics
- Monitoring and Validation Tools